

# Machine Learning for Customer Churn Prediction

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## 1 Executive Summary

Customer Churn is a problem about customer discontinue a company's services or products. Machine learning techniques can be applied with purpose of predict if a customer has probability of be churn, this information can help the company to take decisions for not lost clients. This work search explore different machine learning techniques to compare and get the best approach to solve this problem.

## 2 Dataset Download and Integrity Verification

### 2.1 Dataset Source and Acquisition

The Telco Customer Churn dataset was obtained from the public Kaggle page published by BlastChar and identified as originating from IBM Sample Data Sets. To reproduce the setup, download the complete `WA_Fn-UseC_-Telco-Customer-Churn.csv` file from <https://www.kaggle.com/datasets/blastchar/telco-customer-churn> and place it at `data/WA_Fn-UseC_-Telco-Customer-Churn.csv`. The Kaggle page does not specify a standard open-source license; this project uses the data for academic learning and analysis and acknowledges its source and original authors.

### 2.2 Dataset Characteristics

The CSV is 977,501 bytes (approximately 977.5 KB) and contains 7,043 customer records and 21 columns: 20 customer and service features, plus the binary `Churn` target. It has 1,869 `Yes` and 5,174 `No` records, indicating class imbalance.

### 2.3 Integrity Verification

The provider does not supply an official checksum. To calculate the SHA-256 hash of the local project copy, run `pythonsrc/w1_supervised_learning/dataset_info/checksum.py` from the repository root. The script reads the CSV under `data/` and prints its checksum:

```
88BE4B93FBE0CC83421AF1C503794C97C342ECA914C1576DB7C276E61D61358A
```

This value is a local reference because Kaggle does not provide an official checksum for comparison. To generate a structured dataset report, run `pythonsrc/w1_supervised_learning/dataset_info/dataset_info.py` from the repository root. It prints JSON with the file size, row and column counts, samples per **Churn** class, each column's data type and missing-value count, and numeric feature ranges. Both scripts use the Telco CSV by default; a different file can be supplied with `--csv`.

### 3 Exploratory Data Analysis (EDA)

This section explore the data, explain what represents each feature and respective type, the proportion in the values of numerical and categorical features, the relation between each feature with the target and visual representation of statistics obtained.

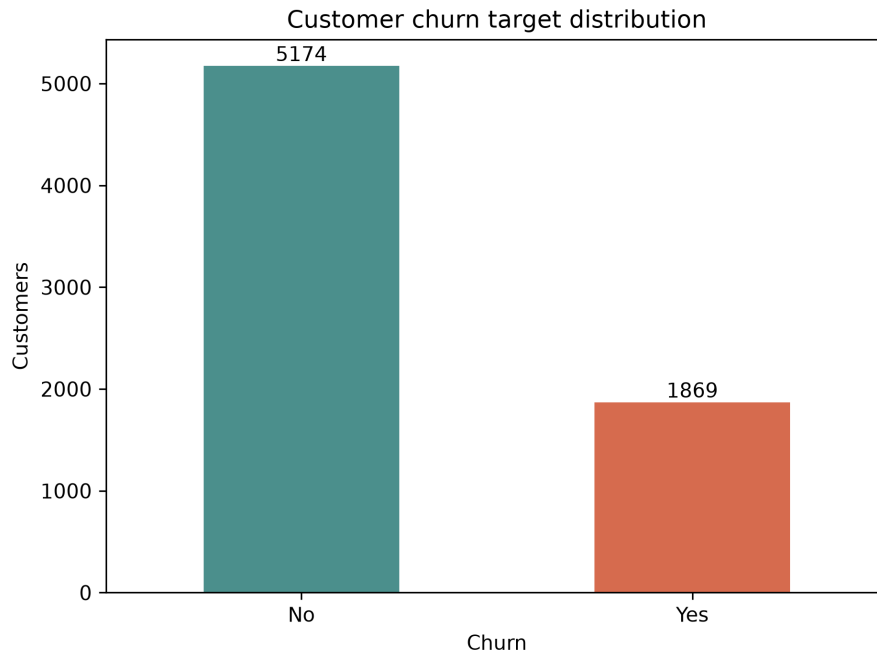
All process in this section can be explore in EDA notebook.

| Feature          | Description                                                                             | Data type |
|------------------|-----------------------------------------------------------------------------------------|-----------|
| customerID       | Unique value that identify a customer                                                   | str       |
| gender           | Determines whether customer is Male or Female                                           | str       |
| SeniorCitizen    | Determines whether a person is citizen or not                                           | int64     |
| Partner          | Determines whether person has partner                                                   | str       |
| Dependents       | Determines whether customer has a dependent                                             | str       |
| tenure           | Number of months has stayed with the company                                            | int64     |
| PhoneService     | Whether customer has phone service                                                      | str       |
| MultipleLines    | Whether the customer has multiple lines, or one, or does not have phone service         | str       |
| InternetService  | Internet service provider (Fiber optic, DSL or no)                                      | str       |
| OnlineSecurity   | Whether customer has online security, or not, or does not have internet service         | str       |
| OnlineBackup     | Whether customer has online backup, or not, or does not have internet service           | str       |
| DeviceProtection | Whether customer has device protection, or not, or does not have internet service       | str       |
| TechSupport      | Whether customer has tech support, or not, or does not have internet service            | str       |
| StreamingTV      | Whether customer has streaming tv, or not, or does not have internet service            | str       |
| StreamingMovies  | Whether customer has streaming movies, or not, or does not have internet service        | str       |
| Contract         | Contract term duration (Month-to-month, One year, Two year)                             | str       |
| PaperlessBilling | Whether customer has paperless billing or not                                           | str       |
| PaymentMethod    | Payment method (Electronic check, Mailed check, Bank transfer (automatic), Credit card) | str       |
| MonthlyCharges   | Value paid by customer monthly                                                          | float64   |
| TotalCharges     | Total value paid by customer                                                            | str       |
| Churn            | Whether the customer churned or not, stopped using the service last month.              | str       |

**Table 1:** Dataset features description

### 3.1 Target Distribution

Of the 7,043 customers, 5,174 (73.46%) stayed and 1,869 (26.54%) churned. The majority-to-minority ratio is 2.77:1, so the target is moderately imbalanced. The class counts are summarized in Table ?? and visualized as a bar chart in the EDA notebook.



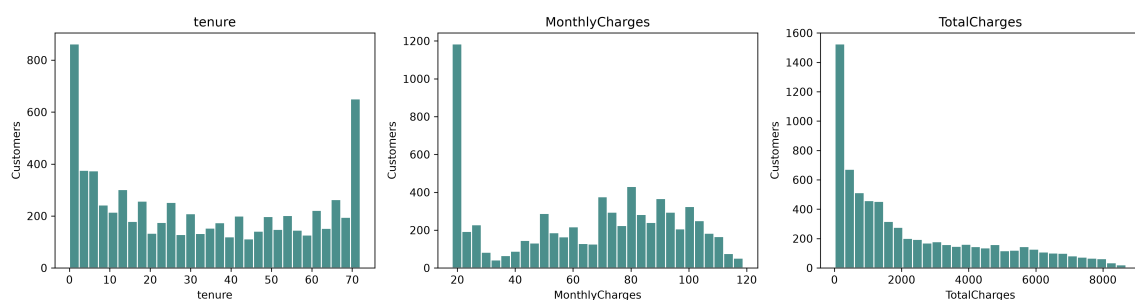
**Figure 1:** Distribution of customers by churn status.

### 3.2 Feature Statistics

Table 2 summarizes the non-binary numeric features: customer tenure and charges. The binary **SeniorCitizen** feature is excluded from these measurements.

| Feature         | N     | Mean     | SD       | Min   | Max      |
|-----------------|-------|----------|----------|-------|----------|
| tenure (months) | 7,043 | 32.37    | 24.56    | 0     | 72       |
| MonthlyCharges  | 7,043 | 64.76    | 30.09    | 18.25 | 118.75   |
| TotalCharges    | 7,032 | 2,283.30 | 2,266.77 | 18.80 | 8,684.80 |

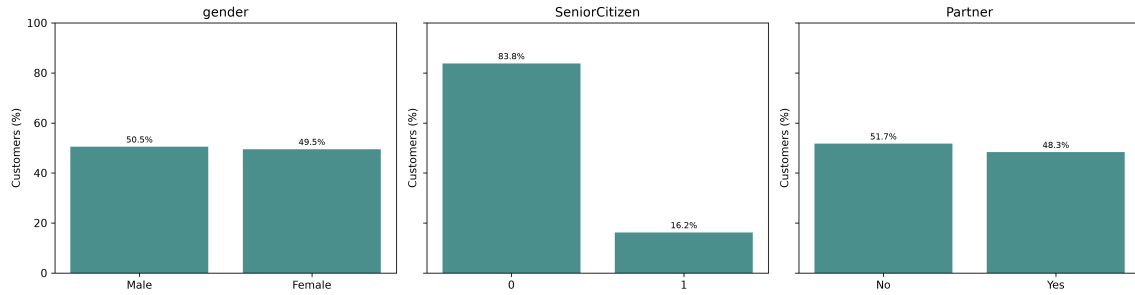
**Table 2:** Summary statistics for numeric features.



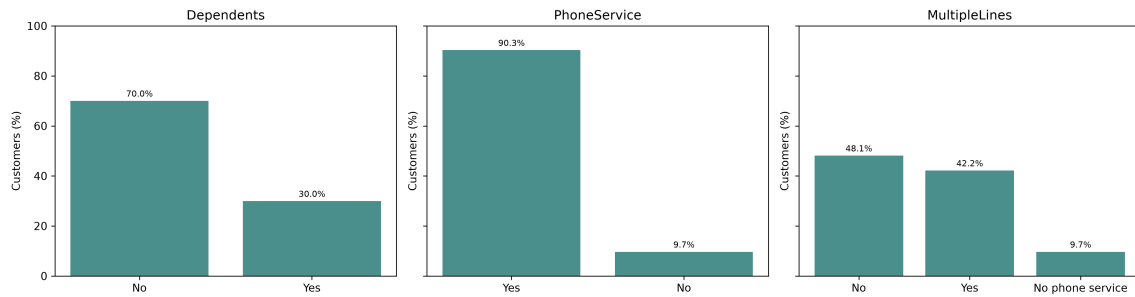
**Figure 2:** Histograms of tenure, monthly charges, and total charges.

### 3.3 Categorical Features

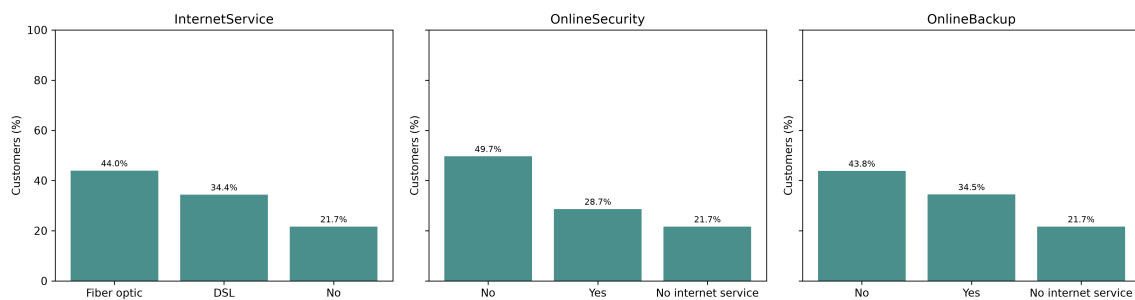
The categorical features describe customer demographics and household status, phone and internet services, service add-ons, streaming, contract and billing arrangements, and payment method. Each bar shows the percentage of all 7,043 customers in that category. The unique identifier `customerID` and target `Churn` are excluded here; the target distribution is shown separately in Figure 1.



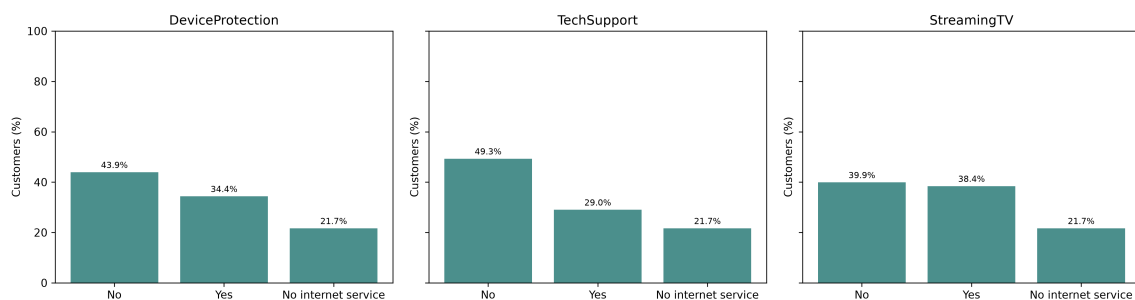
**Figure 3:** Distributions of customer profile features.



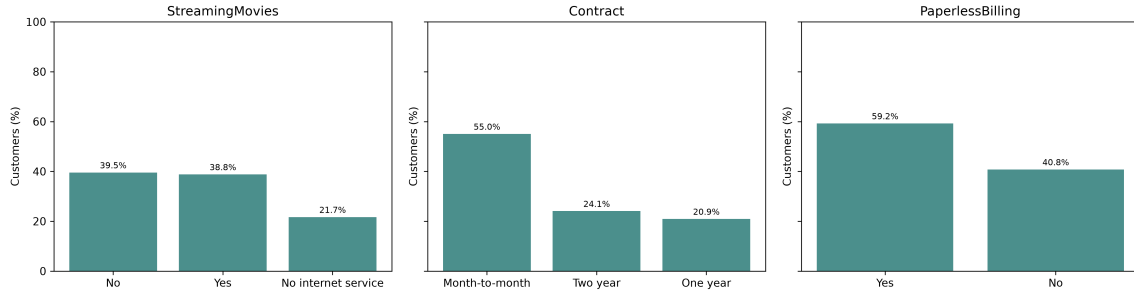
**Figure 4:** Distributions of household and phone service features.



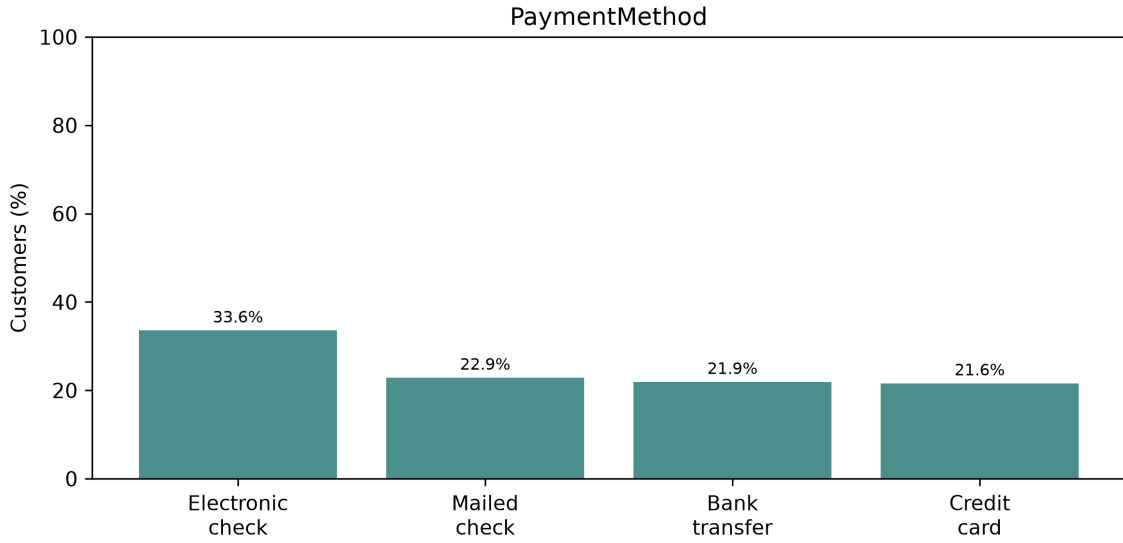
**Figure 5:** Distributions of internet service and online service features.



**Figure 6:** Distributions of device protection, technical support, and streaming TV.



**Figure 7:** Distributions of streaming movies, contract, and paperless billing.



**Figure 8:** Distribution of customer payment methods.

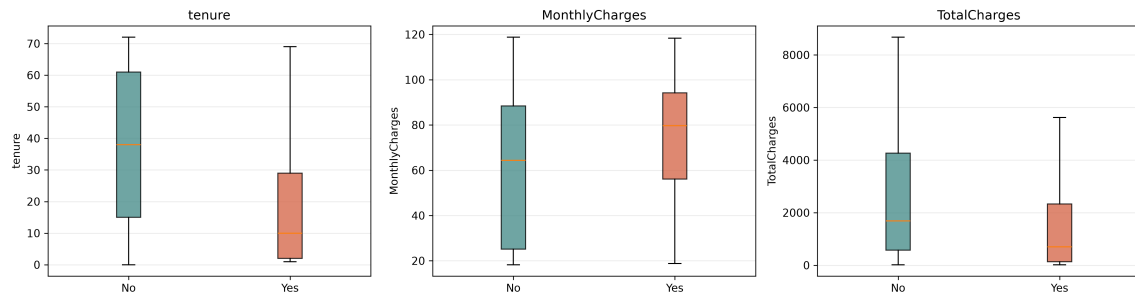
### 3.4 Data Quality

The CSV contains no parsed missing cells or duplicate rows, and all 7,043 customer IDs are unique. However, 11 **TotalCharges** values are blank strings (0.16% of records). Is a little portion unusable. There are 7,032 values equal (99.84% of records )

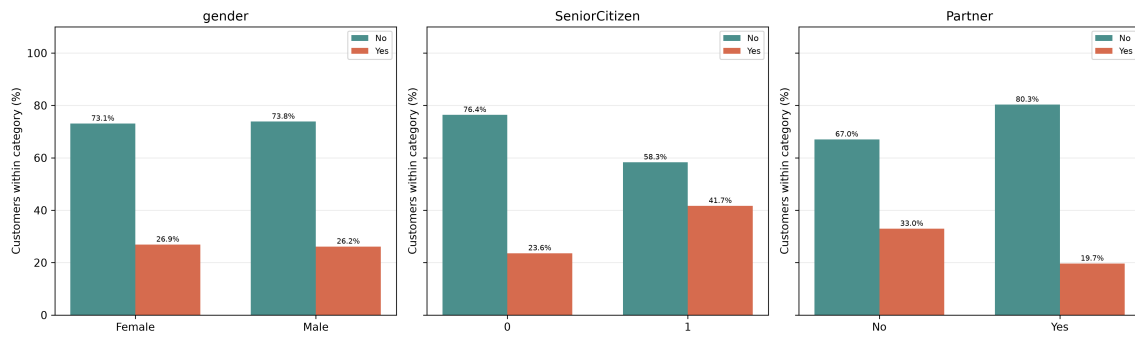
### 3.5 Feature Relationships

These charts compare each feature with the churn target to identify differences between customers who stayed and those who left. The numerical-feature boxplots compare the distributions of tenure, monthly charges, and total charges across the two target classes. For categorical features, each pair of bars shows the percentage of customers within a category who stayed or churned; the two percentages sum to 100% for each category. These are descriptive associations and do not establish causation.

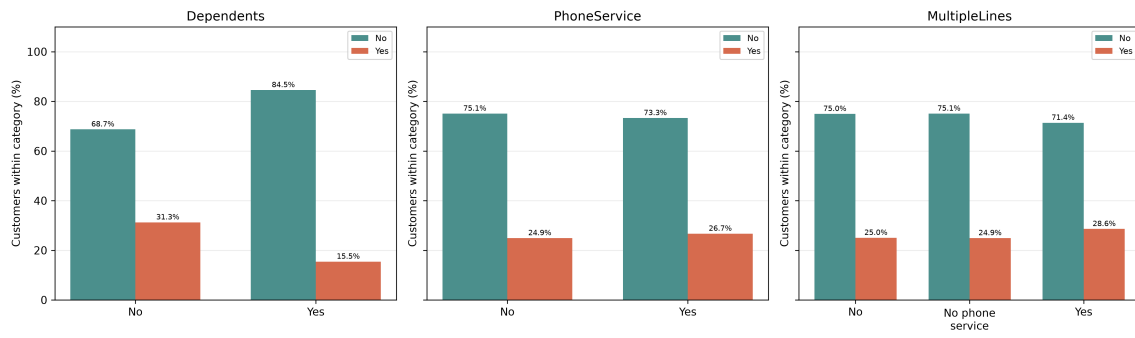
The numerical comparisons show that churned customers have a shorter median tenure (10 months versus 38) and higher mean monthly charges (74.44 versus 61.27). Among categorical features, churn is more common for month-to-month contracts (42.7%) than one-year (11.3%) or two-year contracts (2.8%), and for fiber-optic internet (41.9%) and electronic-check payments (45.3%).



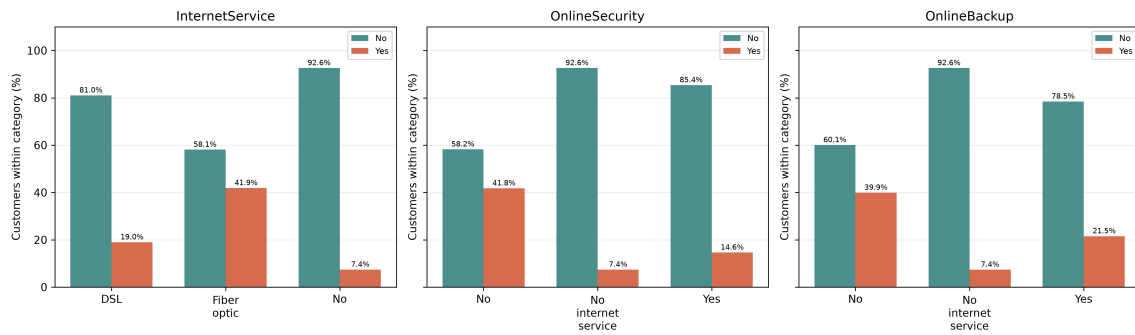
**Figure 9:** Numeric feature distributions by churn status.



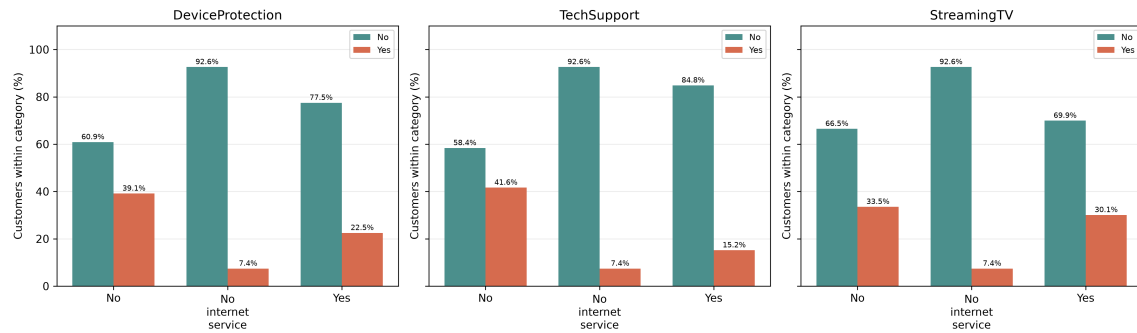
**Figure 10:** Churn rates by customer profile features.



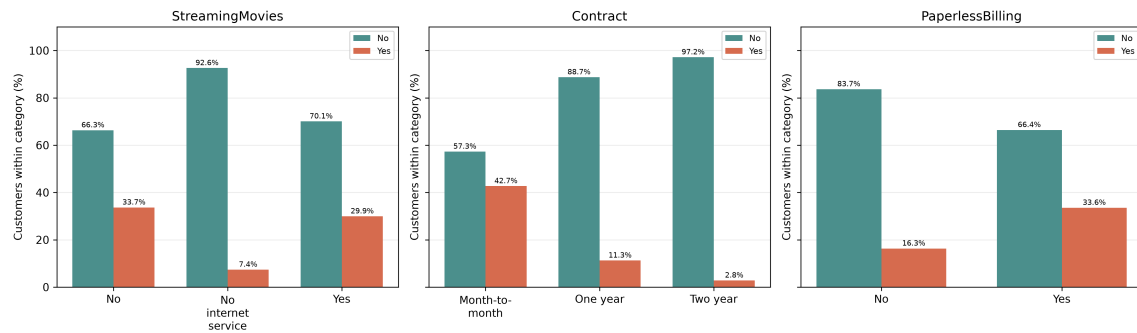
**Figure 11:** Churn rates by household and phone features.



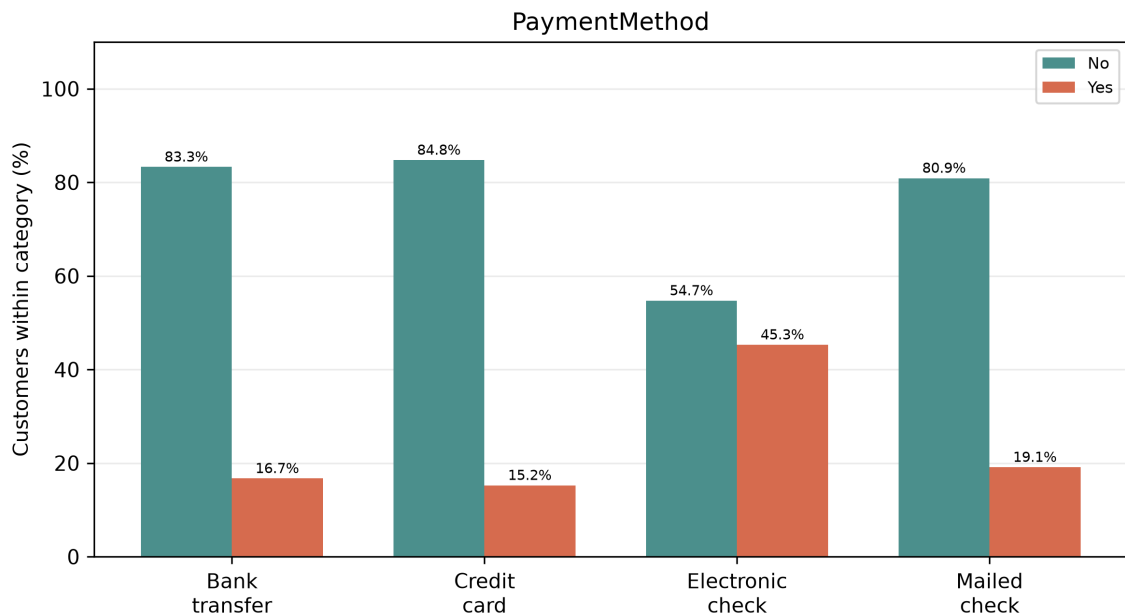
**Figure 12:** Churn rates by internet and online security features.



**Figure 13:** Churn rates by device protection, tech support, and streaming TV.



**Figure 14:** Churn rates by streaming movies, contract, and billing features.



**Figure 15:** Churn rates by payment method.

## 4 Preprocessing Pipeline:

## 5 Models

## 6 Results and Discussion

Explain the experiments you performed to demonstrate that your solution works. Summarize your unit tests in a paragraph, mentioning their philosophy, quantity, and results. Also, discuss any integration and acceptance tests to address software quality and user expectations. If possible, compare your results with other solutions using metrics. Use tables to summarize test definitions and results, and charts (such as box-plots or time series) for comparisons, depending on your project. Highlight the most important findings. Every figure and table must have a descriptive caption and be referenced in the text.

## 7 Conclusion

Write a couple of paragraphs to summarize your work, results, and achievements. Clearly state why your work was successful and how it helped to solve the problem. End with suggestions for future work or open questions that remain

## References

- [1] Rummery, G. A., & Niranjan, M. (1994). On-line Q-learning using connectionist systems. *Technical Report*, Cambridge University Engineering Department.